

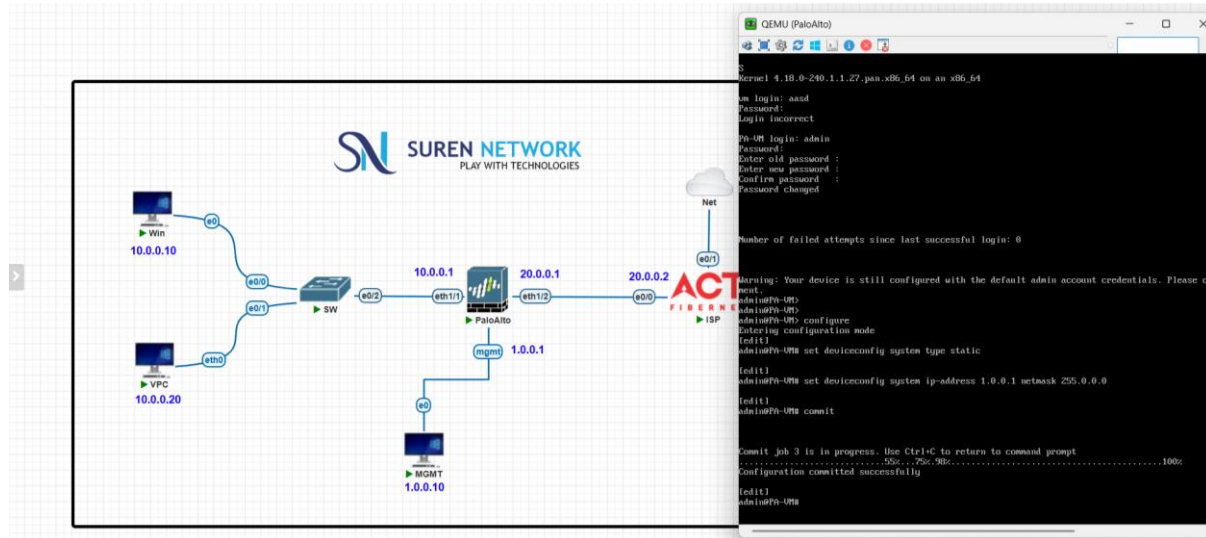
LAB 1 INITIAL CONFIGURATION

GURUCHANDHRAN M

TASK 1: IP ASSIGNMENT

PALO IP: 1.0.0.1

SUBNET MASK: 255.0.0.0



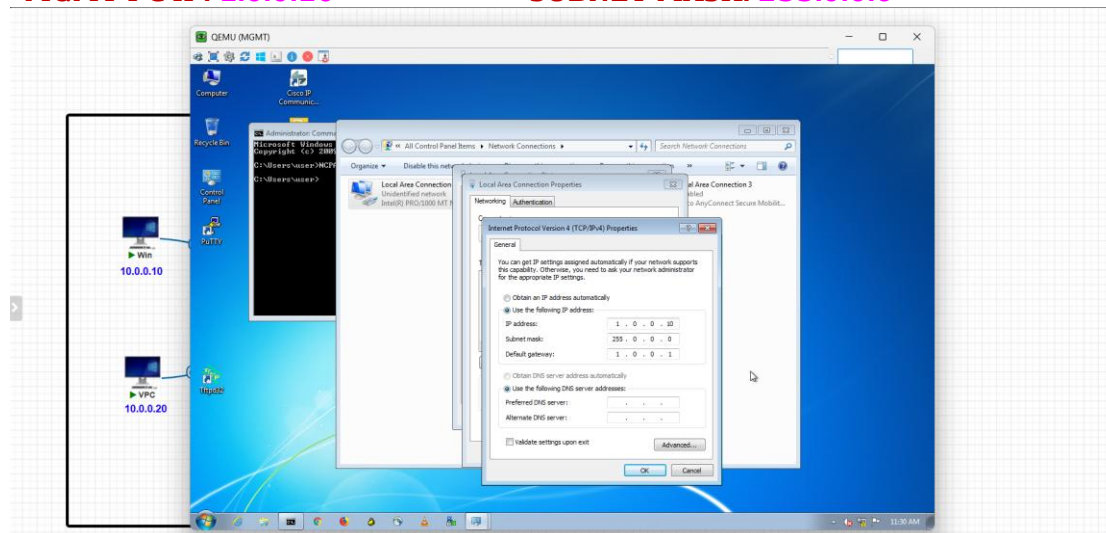
The network diagram shows a topology with a central Palo Alto firewall connected to a switch (SW) and a management PC (MGMT). The switch is connected to two PCs: Win (10.0.0.10) and VPC (10.0.0.20). The Palo Alto firewall has interfaces e0/0, e0/1, e0/2, e0/3, and mgmt. The mgmt interface is assigned IP 1.0.0.1. The switch has interfaces e0/0, e0/1, e0/2, and e0/3. The Win PC has interface e0 and IP 10.0.0.10. The VPC PC has interface e0 and IP 10.0.0.20. The MGMT PC has interface e0 and IP 1.0.0.10. The Palo Alto firewall is connected to the switch via e0/2 and to the MGMT PC via e0/3. The Palo Alto firewall is also connected to an external network (Net) via e0/0 and to an ISP via e0/1.

```
admin@PaloAlto:~$ ssh -p 2200 PaloAlto
Warning: Your device is still configured with the default admin account credentials. Please change the password.
admin(PaloAlto-UTM)
admin(PaloAlto-UTM) configure
admin(PaloAlto-UTM) # configure configuration mode
[edit]
admin(PaloAlto-UTM) set deviceconfig system type static
[edit]
admin(PaloAlto-UTM) set deviceconfig system ip-address 1.0.0.1 netmask 255.0.0.0
[edit]
admin(PaloAlto-UTM) commit
Commit job 3 is in progress. Use Ctrl+C to return to command prompt
Configuration committed successfully
[edit]
admin(PaloAlto-UTM)
```

TASK 2: MANAGEMENT PC IP ASSIGNMENT

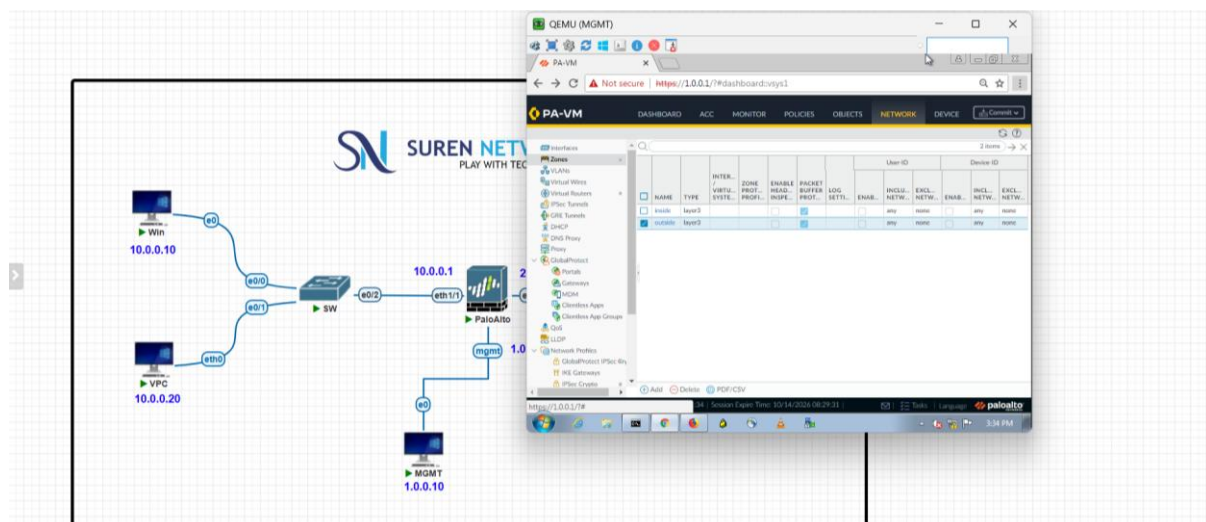
MGMT PC IP: 1.0.0.10

SUBNET MASK: 255.0.0.0



The screenshot shows the Windows 7 desktop of the Management PC. The 'Local Area Connections' window is open, showing the 'Local Area Connection 3' properties. The 'Internet Protocol Version 4 (TCP/IPv4) Properties' window is also open, showing the 'General' tab. The IP address is set to 1.0.0.10, the subnet mask is 255.0.0.0, and the default gateway is 1.0.0.1.

TASK 3: GUI - PALOALTO FIREWALL



The screenshot shows the Palo Alto GUI. The 'PA-VM' window is open, displaying the 'DASHBOARD' view. The 'Zones' tab is selected, showing a table of zones. The table has columns for NAME, TYPE, INTER. VTYPE, ZONE, ENABLE, BUFFER, LOG, LOG, ENAB, ENCL, EXCL, ENAB, EXCL, and NAME. The table contains two rows: 'Inside' and 'Outside'.

NAME	TYPE	INTER. VTYPE	ZONE	ENABLE	BUFFER	LOG	LOG	ENAB	ENCL	EXCL	ENAB	EXCL	NAME
Inside	Host	Host	Host	Enable	1024	1024	1024	1024	any	none	any	none	any
Outside	Host	Host	Host	Enable	1024	1024	1024	1024	any	none	any	none	any

TASK 4: SETTING UP INTERFACES (IP Assignments, Default Virtual Routers)

The screenshot displays the SUREN NETWORK PA-VM interface. On the left, a network topology diagram shows a central switch (SW) connected to a Palo Alto firewall (PA-VM) and a MGMT host. The PA-VM is connected to a VPC (Virtual Private Cloud) and a Win (Windows) host. The PA-VM is configured with IP addresses 10.0.0.10, 10.0.0.20, and 10.0.0.1. The MGMT host is configured with IP address 1.0.0.10. The PA-VM is connected to the VPC and Win hosts via interfaces eth0, eth1, and eth2. The PA-VM is also connected to the MGMT host via interface eth3. The PA-VM is configured with IP addresses 10.0.0.1, 10.0.0.10, 10.0.0.20, and 1.0.0.10. The PA-VM is connected to the VPC and Win hosts via interfaces eth0, eth1, and eth2. The PA-VM is also connected to the MGMT host via interface eth3.

The right pane shows the 'Interfaces' configuration page. The table below lists the configured interfaces:

INTERFACE	INTERFACE TYPE	MANAGEMENT PROFILE	LINK STATE	IP ADDRESS	VIRTUAL ROUTER	TAG	STATUS
ethernet1/1	Layer3		Up	10.0.0.1/8	default		Unassigned
ethernet1/2	Layer3		Up	10.0.0.1/8	default		Unassigned
ethernet1/3	Layer3		Up	10.0.0.1/8	default		Unassigned
ethernet1/4	Layer3		Up	10.0.0.1/8	default		Unassigned
ethernet1/5	Layer3		Up	10.0.0.1/8	default		Unassigned
ethernet1/6	Layer3		Up	10.0.0.1/8	default		Unassigned
ethernet1/7	Layer3		Up	10.0.0.1/8	default		Unassigned
ethernet1/8	Layer3		Up	10.0.0.1/8	default		Unassigned
ethernet1/9	Layer3		Up	10.0.0.1/8	default		Unassigned
ethernet1/10	Layer3		Up	10.0.0.1/8	default		Unassigned

TASK 5: ASSIGNING DEFAULT ROUTE FACING TO ISP

The screenshot displays the SUREN NETWORK PA-VM interface. On the left, a network topology diagram shows a central switch (SW) connected to a Palo Alto firewall (PA-VM) and a MGMT host. The PA-VM is connected to a VPC (Virtual Private Cloud) and a Win (Windows) host. The PA-VM is configured with IP addresses 10.0.0.10, 10.0.0.20, and 10.0.0.1. The MGMT host is configured with IP address 1.0.0.10. The PA-VM is connected to the VPC and Win hosts via interfaces eth0, eth1, and eth2. The PA-VM is also connected to the MGMT host via interface eth3. The PA-VM is configured with IP addresses 10.0.0.1, 10.0.0.10, 10.0.0.20, and 1.0.0.10. The PA-VM is connected to the VPC and Win hosts via interfaces eth0, eth1, and eth2. The PA-VM is also connected to the MGMT host via interface eth3.

The right pane shows the 'Virtual Router - default' configuration page. The table below lists the configured static routes:

NAME	DESTINA.	INTERFACE	TYPE	VALUE	ADMIN. DISTANCE	METRIC	BFD	ROUTE TABLE
Default	0.0.0.0/0	ethernet1/1	ip-destination	20.0.0.2	default	10	None	uninstall

TASK 6: CREATING NETWORK OBJECT (LAN-SW) FOR LAN NETWORK

The screenshot displays the SUREN NETWORK PA-VM interface. On the left, a network topology diagram shows a central switch (SW) connected to a Palo Alto firewall (PA-VM) and a MGMT host. The PA-VM is connected to a VPC (Virtual Private Cloud) and a Win (Windows) host. The PA-VM is configured with IP addresses 10.0.0.10, 10.0.0.20, and 10.0.0.1. The MGMT host is configured with IP address 1.0.0.10. The PA-VM is connected to the VPC and Win hosts via interfaces eth0, eth1, and eth2. The PA-VM is also connected to the MGMT host via interface eth3. The PA-VM is configured with IP addresses 10.0.0.1, 10.0.0.10, 10.0.0.20, and 1.0.0.10. The PA-VM is connected to the VPC and Win hosts via interfaces eth0, eth1, and eth2. The PA-VM is also connected to the MGMT host via interface eth3.

The right pane shows the 'Address' configuration page. The table below lists the configured address objects:

NAME	LOCATION	TYPE	ADDRESS	TAGS
LAN-SW		IP Network	10.0.0.0/8	

TASK 7: CREATING SERVICE OBJECT FOR DNS

The network diagram shows a topology with a central switch (SW) connected to a Palo Alto VM. The Palo Alto VM has interfaces eth1/1 (10.0.0.1) and eth1/2 (20.0.0.1). A MGMT interface (1.0.0.1) is also shown. The Palo Alto VM is connected to a MGMT host (1.0.0.10). The Palo Alto VM is configured with a Service object for DNS.

The Palo Alto VM configuration window shows the Service object configuration:

- Name: DNS
- Description:
- Protocol: TCP, UDP
- Destination Port: 53
- Source Port: Any
- Session Timeout: Inherit from application
- Tags:

TASK 8: DRAFTING SECURITY POLICIES TO REACH INTERNET

The network diagram is the same as in Task 7. The Palo Alto VM is configured with a Security Policy to allow traffic to the internet.

The Palo Alto VM configuration window shows the Security Policy configuration:

NAME	TAGS	TYPE	ZONE	ADDRESS	USER	SERVICE	ZONE
1. Internet		Permit	Inside	Any	Any	Any	Outside
2. Internet		Deny	Inside	Any	Any	Any	Outside
3. Internet		Deny	Inside	Any	Any	Any	Outside

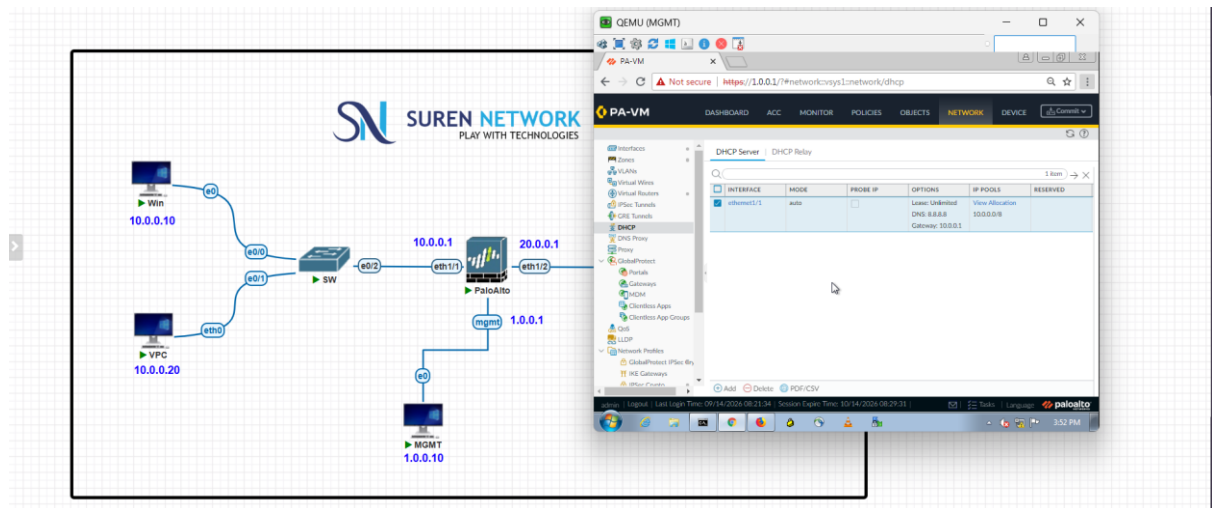
TASK 9: DRAFTING NAT POLICIES TO REACH INTERNET

The network diagram is the same as in Task 7. The Palo Alto VM is configured with a NAT Policy to allow traffic to the internet.

The Palo Alto VM configuration window shows the NAT Policy configuration:

NAME	TAGS	SOURCE ZONE	DESTINATION ZONE	DESTINATION INTERFACE	SOURCE ADDRESS
1. NAT		Inside	Outside	ethernet1/2	10.0.0.10

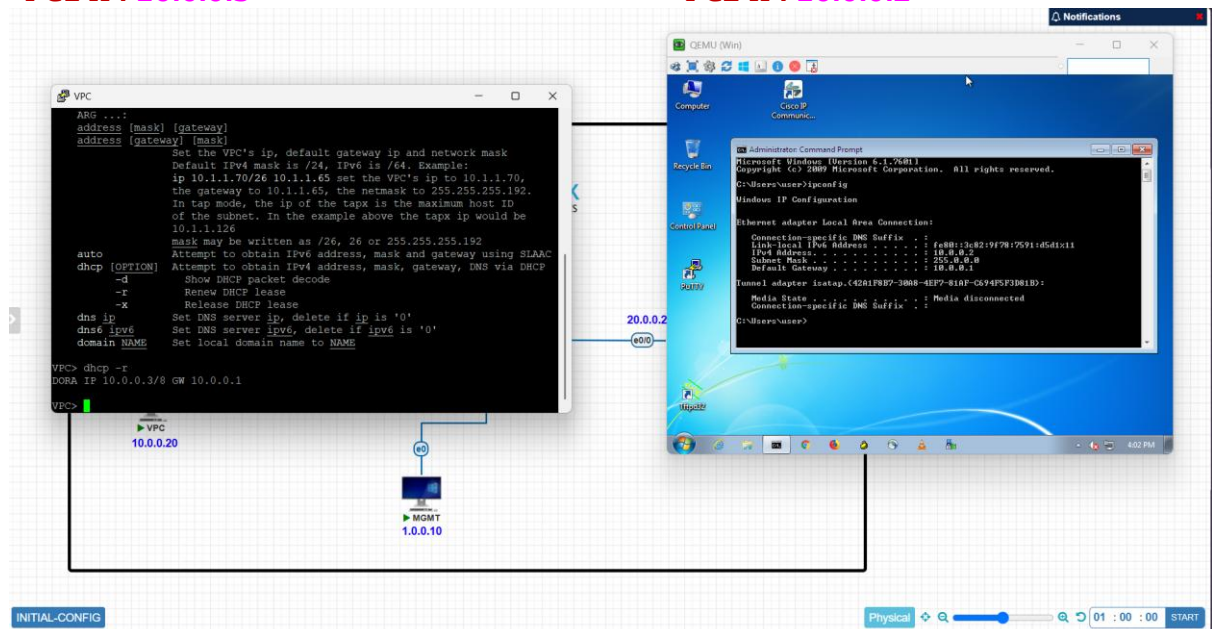
TASK 10: ENABLING DHCP FOR LAN INTERFACE



TASK 11: PC IP ASSIGNMENT

PC1 IP: 10.0.0.3

PC2 IP: 10.0.0.2



TASK 12: ACCESSING INTERNET FROM THE LAN PC

